

1 RelCalc report for spin system: NH

Spins: N₁:+ $\frac{1}{2}$ H₂:+ $\frac{1}{2}$

Interactions:

Dipolar between H₂ and N₁ $\beta = \beta$ $\alpha = 0$ distance R_d constant $d = \frac{\mu_0 \hbar \gamma_H \gamma_N}{4\pi (R_d)^3}$,
local motion: static

CSA for N₁ $\beta = \alpha = 0$ constant $c_A = \gamma_N B_0 \Delta \delta_{sym}/3$, $\Delta \delta_{sym} = \delta_{zz} - \delta_{yy}$,
local motion: static

CSA for H₂ $\beta = 0$ $\alpha = 0$ constant $c_X = \gamma_H B_0 \Delta \delta_{sym}/3$, $\Delta \delta_{sym} = \delta_{zz} - \delta_{yy}$,
local motion: static

Symbolic factors:

Legendre polynomials: β : (rank 2; P_q) : (rank 2; Q_q).

Macromolecular terms only: False

Spectral density functions: Axially symmetric tumbling, static in the molecular frame (AX-STAT)

$$J(q, \omega) = \frac{\tau(q)}{1 + \omega^2 \tau(q)^2}$$

$$L(\beta_1, \alpha_1, \beta_2, \alpha_2, \omega) = \sum_q^2 y_q^2 P_q(\cos \beta_1') P_q(\cos \beta_2') \cos(q(\alpha_1' - \alpha_2')) J(q, \omega)$$

$L(\beta_1, \beta_2, \omega)$ means $\alpha_1 = \alpha_2$

$L(\beta, \omega)$ means $\alpha_1 = \alpha_2$ and $\beta_1 = \beta_2$

For AX-STAT assume molecular frame is pre-aligned with the diffusion frame. Note:

adjust the molecular frame angles from $\beta_{1/2}$ and $\alpha_{1/2}$ into the diffusion frame

(θ_O, ϕ_O) via (not done automatically). Function `p0,cosTheta,sinPhi,cosPhi=GetEffectiveAngles(theta,phi)` is provided:

$$\cos \beta' = (\cos(\beta) \cos(\theta_O) - \cos(\alpha + \phi_O) \sin(\beta) \sin(\theta_O))^2$$

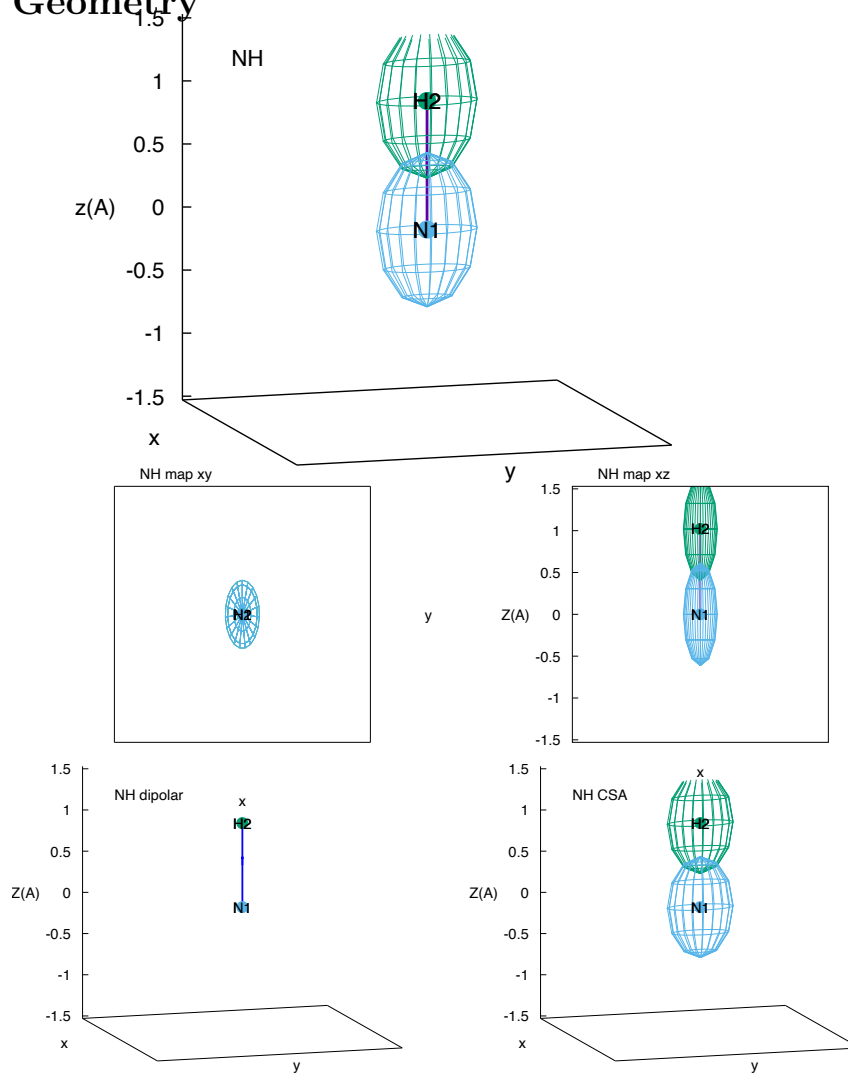
$$\sin \alpha' = \sin(\beta) \left(\frac{1}{2} \sin(\theta_O) \sin(2(\phi_O + \alpha)) \sin(\beta) - \cos(\theta_O) \sin(\phi_O + \alpha) \cos(\beta) \right)$$

$$\cos \alpha' = \frac{1}{4} \left(\sin(2\theta_O) (3 \cos(\beta)^2 - 1) + 4 (2 \cos(\theta_O)^2 - 1) \cos(\phi_O + \alpha) \sin(\beta) \cos(\beta) - 2 \sin(\theta_O) \cos(\theta_O) \cos(2(\phi_O + \alpha)) \right)$$

Timescale:

$$\frac{1}{\tau(q)} = \frac{1}{\tau_c} + \frac{q^2}{6} \left(\frac{1}{\tau_{\parallel}} - \frac{1}{\tau_c} \right)$$

2 Geometry



The geometry of the NH spin system evaluated for current parameters. The AX dipolar interaction from the first spin is used to calculate the geometry. Arrows on the dipolar interactions should indicate direction. Otherwise indicates an error (other than 'ext' interactions). Axially symmetric tensoral components are represented by an ellipse. Units are Angstroms.

3 Relaxation rates

Rate for $N1p$ on $N1p$:

$$R_{N1p,N1p} = +c_A^2 \left[+\frac{4}{5}L(,0) + \frac{3}{5}L(,\omega_N) \right] + d^2 \left[+\frac{1}{5}L(\beta,0) + \frac{1}{20}L(\beta,\omega_H - \omega_N) + \frac{3}{20}L(\beta,\omega_N) + \frac{3}{10}L(\beta,\omega_H) + \frac{3}{10}L(\beta,\omega_H + \omega_N) \right] \quad (1)$$

Rate for $N1z$ on $N1z$:

$$R_{N1z,N1z} = +c_A^2 \left[+\frac{6}{5}L(,\omega_N) \right] + d^2 \left[+\frac{1}{10}L(\beta,\omega_H - \omega_N) + \frac{3}{10}L(\beta,\omega_N) + \frac{3}{5}L(\beta,\omega_H + \omega_N) \right] \quad (2)$$

Rate for $H2z$ on $N1z$:

$$R_{N1z,H2z} = +d^2 \left[-\frac{1}{10}L(\beta,\omega_H - \omega_N) + \frac{3}{5}L(\beta,\omega_H + \omega_N) \right] \quad (3)$$

Rate for $N1z$ on $H2z$:

$$R_{H2z,N1z} = +d^2 \left[-\frac{1}{10}L(\beta,\omega_H - \omega_N) + \frac{3}{5}L(\beta,\omega_H + \omega_N) \right] \quad (4)$$

Rate for $H2z$ on $H2z$:

$$R_{H2z,H2z} = +c_X^2 \left[+\frac{6}{5}L(0,\omega_H) \right] + d^2 \left[+\frac{1}{10}L(\beta,\omega_H - \omega_N) + \frac{3}{10}L(\beta,\omega_H) + \frac{3}{5}L(\beta,\omega_H + \omega_N) \right] \quad (5)$$

Rate calculation time: 0:00:00.000633

4 Parameter index

Key	Value	Description
d:	(1.02e-10, 'N', 'H')	dipolar between H ₂ and N ₁ (distance+nuclei)
beta:	0.00°	angle for Legendre polynomials: P
betaC:	0.00°	angle for Legendre polynomials: Q
cX:	(10.0, 'H')	CSA for H ₂ (ppm+nucleus)
cA:	(-160.0, 'N')	CSA for N ₁ (ppm+nucleus)

5 Plots at specified values

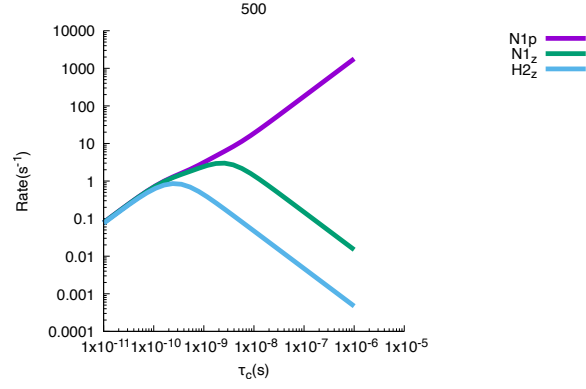


Figure 1: Relaxation rates versus correlation time τ_c for spin system NH. macro=False. ν_0 500.00 MHz. AXIAL DIFFUSION: Rotation into diffusion frame, θ_O 0.25π , ϕ_O 1.00π , shape factor, $1.00\text{e}+01$ ns. $\tau_{\parallel}$ $4.00\text{e}+01$ ns. $\beta=0.00^\circ$ (P rank 2). $=0.00^\circ$ (Q rank 2). Dipolar d : (γ_N, γ_H) , $R_d=1.02\text{\AA}$, $d=7.21\text{e}+04$ rad s⁻². CSA: c_A (γ_N) -160.00ppm, $c_A=1.70\text{e}+04$ rad s⁻². CSA: c_X (γ_H) 10.00ppm, $c_X=1.05\text{e}+04$ rad s⁻².